FORECASTING COVID-19 WITH TEMPORAL HIERARCHIES AND ENSEMBLE METHODS

A Thesis Presented

by

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ABSTRACT

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Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasts of COVID-19 incident hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF's unique approach has shown substantial accuracy improvements in a diverse range of applications, such as operations management and emergency room admission predictions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and weighted ensembles of the THieF models. The models are validated using weighted interval score (WIS), and the top-performing subset is compared to several benchmark models. These models include simple SARIMA models, a naive baseline model, and a median ensemble of operational incident hospitalization models from the US COVID-19 Forecast Hub. The THieF models and THieF ensembles demonstrate improvements in WIS and MAE, as well as competitive prediction interval coverage, over the benchmark models for both the validation and testing phases. These accuracy improvements suggest the THieF methodology may serve as a promising new technique for infectious disease forecasting that makes better predictions in real-time.

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CHAPTER 2

METHODS

2.1 Surveillance data

This analysis uses confirmed hospital admissions from HealthData.gov from the COVID-19 Reported Patient Impact and Hospital Capacity by State Time Series. This dataset reports state-aggregated daily incident hospitalizations starting on January 1, 2020 for U.S. states and territories. The data come from three primary sources: HHSTeleTracking, state/territorial health departments which report to HHS Protect, and National Health Care Safety Network (prior to July 15, 2020) [7].

On December 1, 2020, the U.S. COVID-19 Forecast Hub designated this dataset as the official ground truth data for COVID-19 incident hospitalizations. (No other sources were designated official ground truth data prior to this date.) [2] This meant that the HealthData.gov data was used to evaluate the performance of real-time incident hospitalization forecasts submitted to the Forecast Hub. Hence, teams were encouraged to use this hospitalization data set to train their models [3]. In the analysis, we also use this dataset as both input training data for our models and truth data against which to perform accuracy evaluations.

Hospitalization reporting was sporadic for the first four months of the pandemic, with consistent, reliable values for major states and territories achieved only in late July 2020. Specifically, reports of daily confirmed hospitalizations for all 50 states, Washington D.C., Puerto Rico, and the U.S. as a whole began on July 27, 2020. Although COVID-19 hospitalization reporting stabilized after that for cases and deaths, hospitalization values are often more reliable due to requirements for Medicare reimbursement. COVID-19 cases have been vastly underreported for the entire duration of the pandemic [5], and this problem has worsened with dwindling testing. Further, truth data collection and reporting for COVID-19 incident cases and deaths has been discontinued by Johns Hopkins University (the official ground truth data source) as of March 10, 2023 [4].

- Note though that hospitalization data is not perfect and values may be updated, especially within the first few weeks and incident hospitalization value is posted. However, models are training on the currently available data to them, which may not reflect the finalized truth value - Truth data is queried through Zoltar, a data repository set up by..., which includes past versions of the truth data so that original values may be obtained. The data used to train the models in this analysis is queried such that it would be the data available as of the date that the retrospective forecast is "made on" so as to be comparable to other models.

2.2 Forecasts

2.2.1 Forecast quantiles

2.2.2 Forecast locations

2.2.3 Forecast horizons

2.3 Date range of analysis

2.4 Comparison models

- COVIDhub-baseline: A naive baseline model created and run by the US COVID-19 Forecast Hub meant to serve as a benchmark model to compare against. This model makes forecasts using the previous day's incident hospitalization value as the point forecast (and 0.5 quantile for the probabilistic forecast) and past changes in daily incidence to calibrate uncertainty in its probabilistic forecasts. Beating the baseline is typically seen as one marker of a useful model [3]

- COVIDhub-4_week_ensemble: A median ensemble of all the submitting models for incident hospitalizations at the COVID-19 Forecast Hub, run by the Hub team that includes forecasts up to 28 day (4 week) ahead horizons. This model produces the same 1 to 14 day (1 to 2 week) ahead horizon forecasts as the COVIDhub-ensemble model, but also forecasts longer horizons. These longer horizons were removed from the main ensemble model due to research showing a sharp decline in accuracy in forecasts predicting beyond 2 week aheads (see Forecast horizons). However, these longer horizons are useful as a comparison for research purposes. Note that this ensemble model included slightly different contributing models based on what models were submitting to the COVID-19 Forecast Hub at a different times (check for specifics). [1], [6]

2.5 Temporal hierarchical forecasting (THieF)

- Add more mathematical detail here - Aggregation: We start with data at a daily scale. Then, based on the aggregation levels of interests for a particular model (e.g. weekly, every two weeks, every four weeks, etc), we induce this sort of artificial aggregation so that we have vectors of data add each of these time scales. Next, forecasts are made using each of these datasets separately. The reason that we are interested in different time scales is that there may be important patterns that are lost add higher time scales amid the noise of lower ones. More often than not, different patterns will be picked up at different aggregation levels, which means we get forecasts for the same future time periods that may not agree. This is where reconciliation comes into play, in which we combine all of the forecasts that we made.

2.6 Model specifications

2.6.1 THieF models

2.6.2 THieF ensemble models

2.6.3 ARIMA and SARIMA models

2.7 Metrics and Evaluation

2.7.1 Validation phase

2.7.2 Testing phase

CHAPTER 3

RESULTS

3.1 Validation phase

3.2 Testing phase

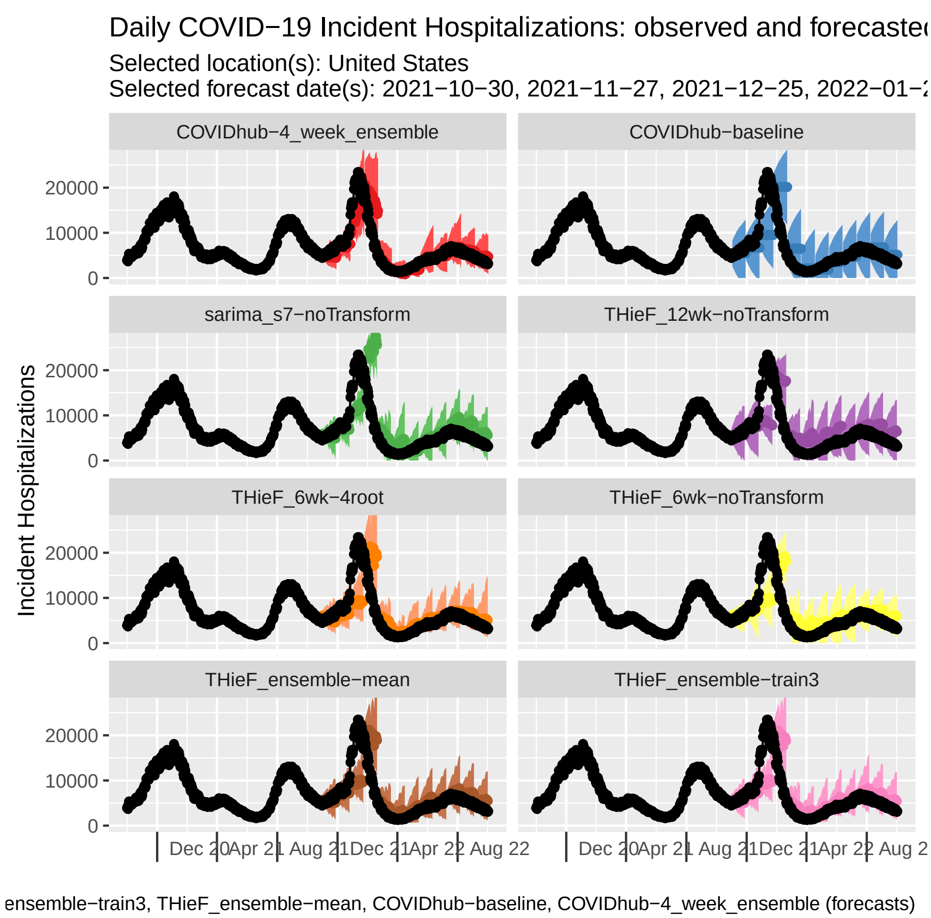

Figure 3.1.

3.2.1 Summary of comparison models

3.2.2 Overall model performance

model	wis	mae	cov_50	cov_95	rwis	rmae
COVIDhub-4_week_ensemble	1401.3	2085.1	0.629	0.912	0.750	0.829
THieF_6wk-4root	1642.5	2358.8	0.400	0.817	0.880	0.938
THieF_ensemble-train3	1687.4	2443.4	0.358	0.829	0.904	0.972
THieF_ensemble-mean	1700.5	2458.9	0.388	0.836	0.911	0.978
THieF_6wk-noTransform	1727.7	2504.8	0.457	0.862	0.925	0.996
COVIDhub-baseline	1867.5	2514.1	0.715	0.852	1.000	1.000
THieF_12wk-noTransform	2074.1	2908.3	0.367	0.864	1.111	1.157
sarima_s7-noTransform	2316.8	3042.4	0.409	0.798	1.241	1.210

Table 3.1. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The "rwis" and "rmae" show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

model	wis	mae	cov_50	cov_95	rwis	rmae
COVIDhub-4_week_ensemble	1401.3	2085.1	0.629	0.912	0.750	0.829
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model	wis	mae	cov_50	cov_95	rwis	rmae
COVIDhub-4_week_ensemble	30.181	44.870	0.644	0.945	0.725	0.790
THieF_6wk-4root	34.549	50.116	0.512	0.895	0.830	0.883
THieF_ensemble-train3	35.625	51.574	0.558	0.906	0.856	0.908
THieF_ensemble-mean	35.914	51.805	0.570	0.906	0.863	0.913
THieF_6wk-noTransform	39.508	56.867	0.546	0.878	0.949	1.002
COVIDhub-baseline	41.629	56.772	0.790	0.957	1.000	1.000
THieF_12wk-noTransform	41.656	60.970	0.510	0.877	1.001	1.074
sarima_s7-noTransform	43.538	58.935	0.596	0.868	1.046	1.038

3.2.3 Model performance by horizon week

3.2.4 Model performance by pandemic wave

3.2.5 Model performance by location

3.2.6 Model performance by forecast week

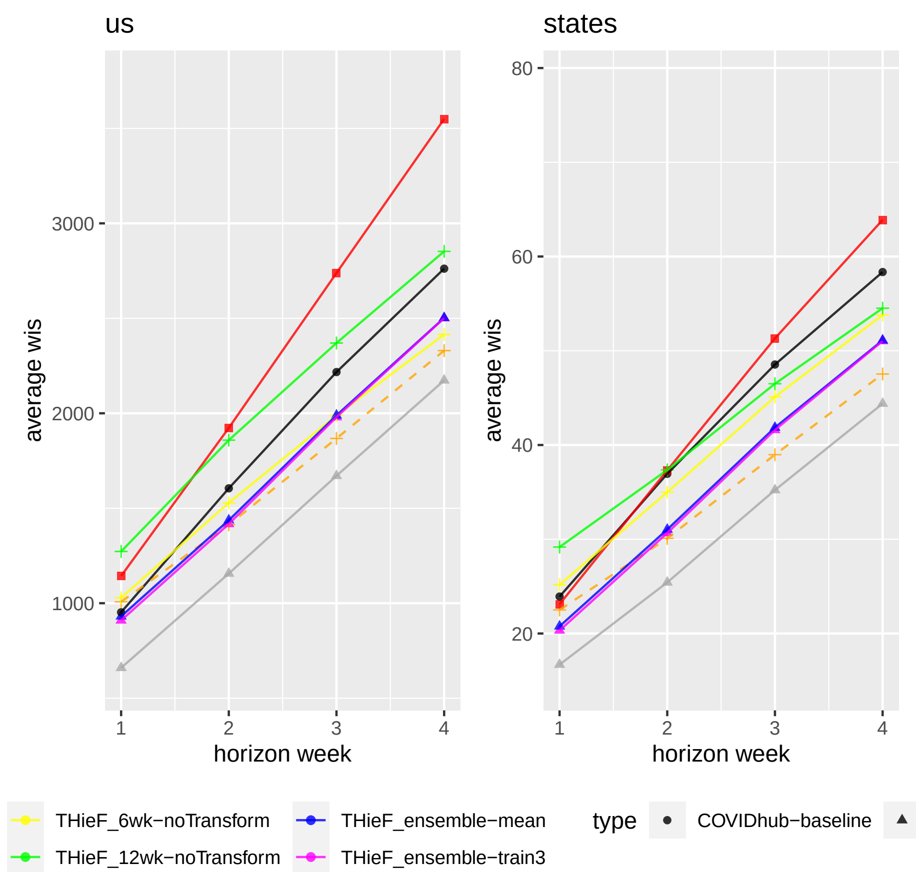

Figure 3.2. Average WIS by the horizon week for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week compared to being aggregated across horizon weeks. All models show worse performance relative to the baseline at the 1-week ahead horizon, likely a result of being overconfident at this horizon. Once again, the THieF-4wk-4root model performs the best for WIS. Plots for average MAE are not shown due to their similarity to the average WIS plots.

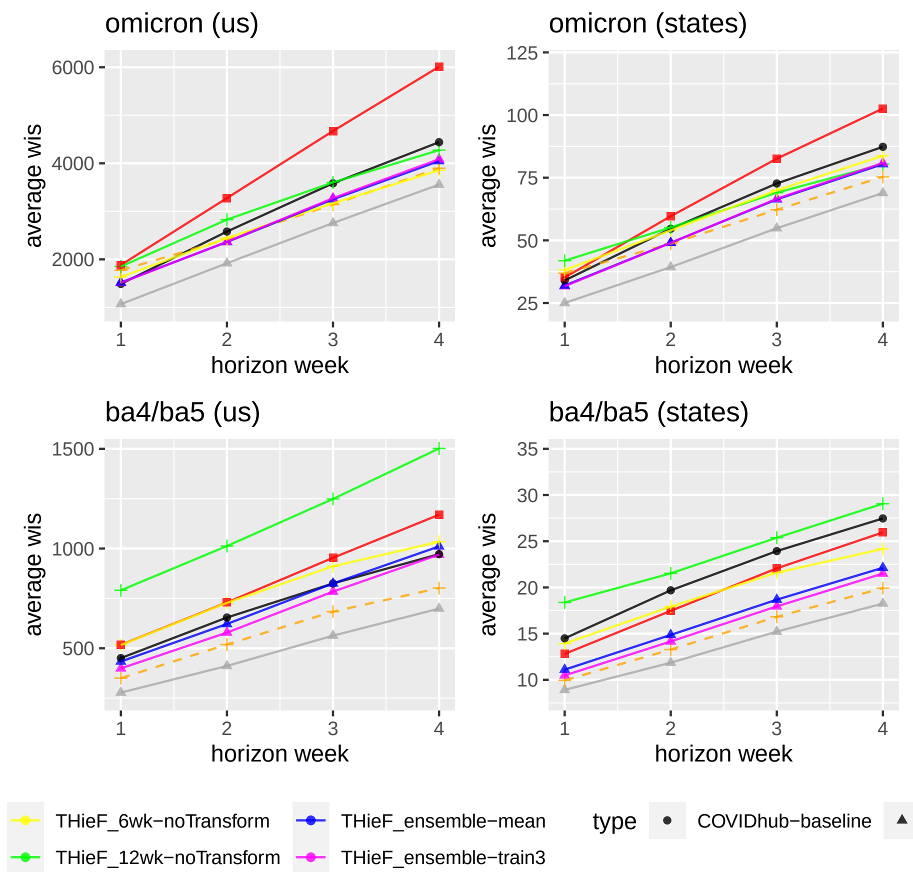

Figure 3.3. Average WIS by the horizon week and pandemic wave for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week and pandemic wave compared to being stratified only across horizon week. Again, all models show worse performance relative to the baseline at the 1-week ahead horizon likely due to being overconfident at this horizon. Model performance is more similar among the top four models during the Alpha wave compared to during the Winter21 wave. Notably, although the THieF-4wk-4root models largely show the lowest WIS, longer horizons during the Winter21 wave saw better performance by the poor-performing THieF-12wk models. Plots for average MAE are not shown due to their similarity to the average WIS plots.

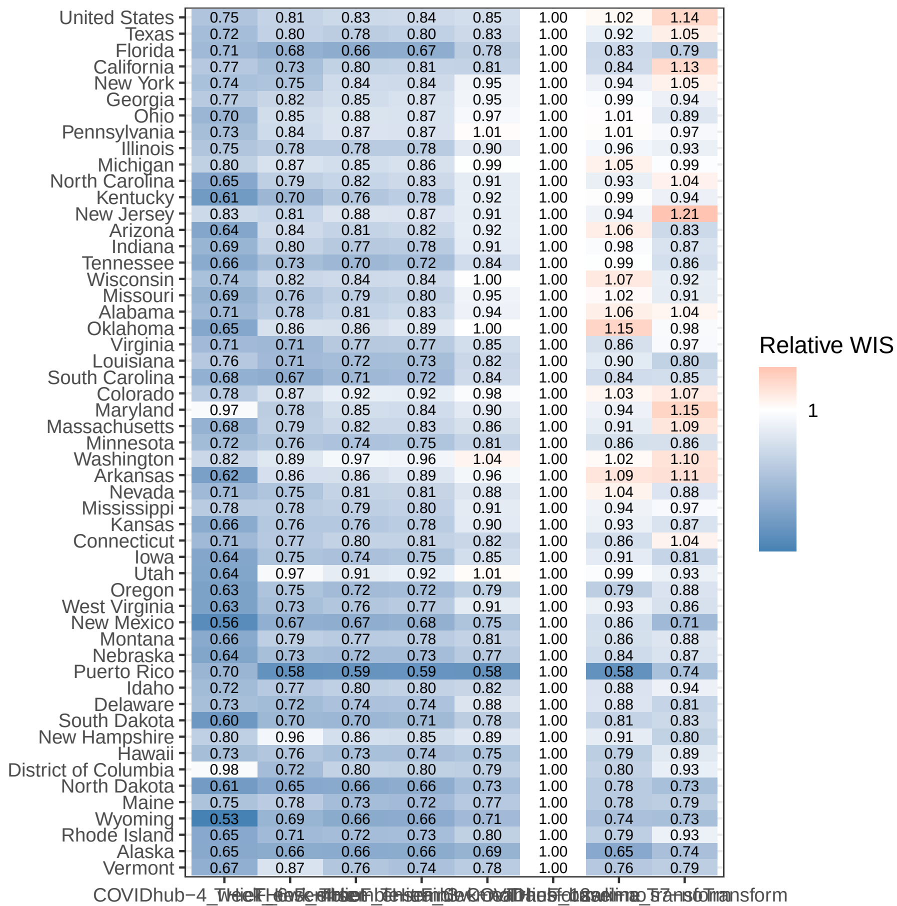

Figure 3.4. Relative WIS by location for each model across all horizons for the entire testing phase. Each square contains the average relative WIS or a model at a particular location, colored based on the relative WIS compare to the baseline. Blue indicates model outperformed the baseline for that location on average, while red indicates the model performed worse than the baseline. Locations are sorted based on cumulative hospitalizations during the testing phase while models are sorted from lowest to highest overall relative WIS, aggregated across states (Table 2). The top three models by relative WIS overall during the testing phase outperformed the baseline in every location.

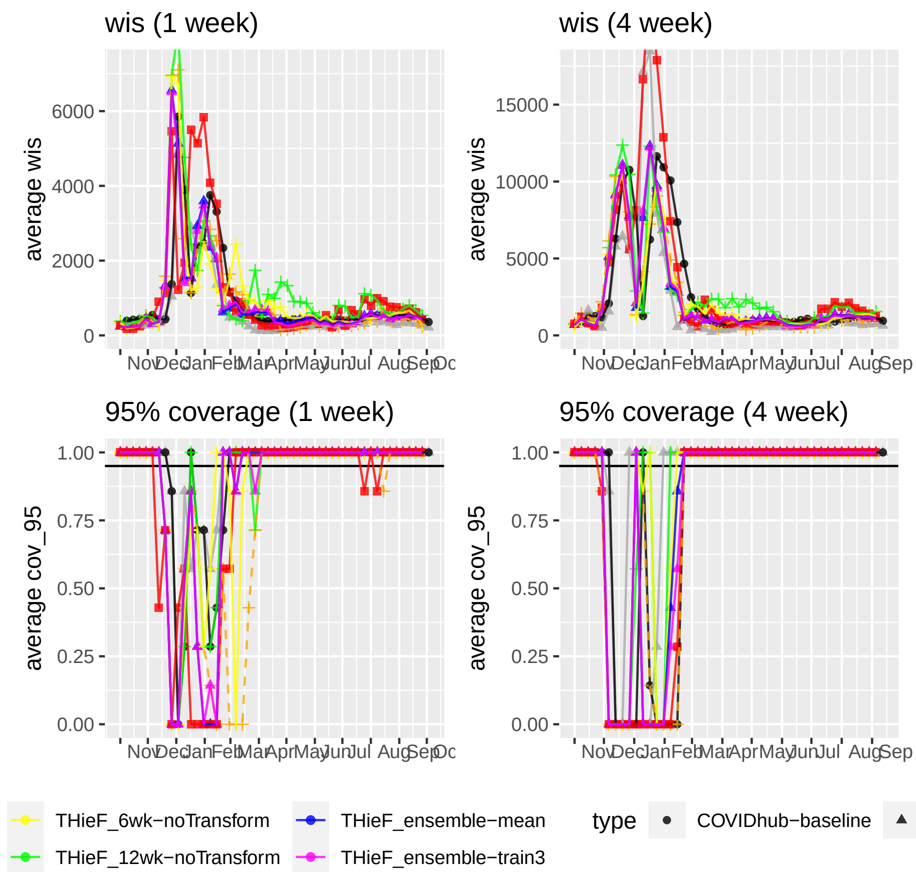

Figure 3.5. Average h-week ahead WIS and 95% PI coverage for each model for the US national level. Model performance differs from week to week: the WIS is the lowest during weeks with stable and low incident hospitalization values. The THieF-4wk models show the best performance with lower or comparable WIS to the baseline during the majority of the training period with especially good performance for 1-week ahead horizons. Meanwhile, the THieF-12wk showed lower or comparable WIS to the baseline during high incidence and changed points, which have been historically found difficult to forecast accurately. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. All THieF models show very variable coverage rates, especially for a 1 week ahead horizon, only the models without a data transformation showing stable coverage starting around February 2021.

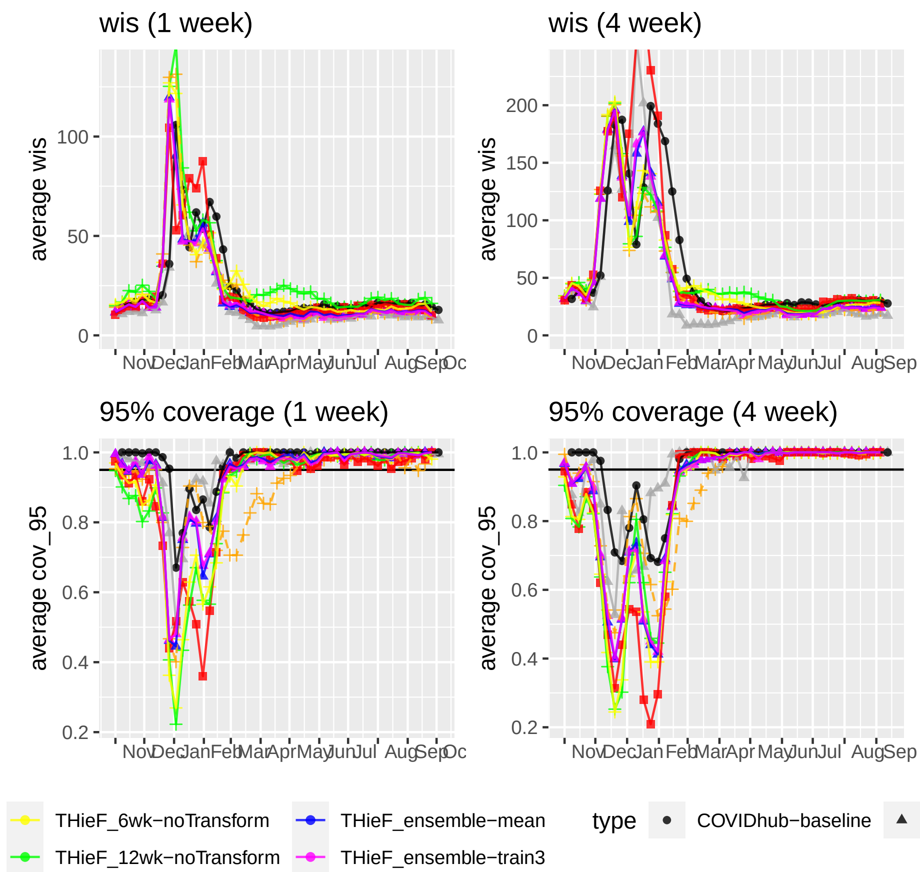

Figure 3.6. Average h-week ahead WIS and 95% PI coverage for each model for the states level. Once again, model performance differs from week to week with the lowest WIS values during weeks with stable and low incident hospitalization values and the THieF-4wk models showing the best performance. However, the THieF-12wk did not show good performance during high incidence and changed points for the states level; instead, THieF-12-4root actually showed some strangely large WIS values during April and May 2021. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. One-week ahead coverage rates generally improved as time went on, the best models gaining stable and fairly good coverage rates starting in February 2021. For the 4-week ahead horizon, models without a data transformation showed the best performance and stable coverage starting around February 2021, though THieF-4wk-4root and THieF-8wk-4root also saw decent and fairly stable coverage rates after the downward Winter 21 change point.

CHAPTER 4

DISCUSSION

APPENDIX: SUPPLEMENTAL MATERIALS

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rm
COVIDhub-4_week_ensemble	1	659.8	966.8	0.682	0.937	0.693	0.7
THieF_ensemble-train3	1	909.6	1348.8	0.366	0.859	0.955	1.0
THieF_ensemble-mean	1	930.9	1372.9	0.381	0.859	0.978	1.0
COVIDhub-baseline	1	952.2	1258.8	0.733	0.916	1.000	1.0
THieF_6wk-4root	1	1007.4	1419.1	0.447	0.787	1.058	1.1
THieF_6wk-noTransform	1	1028.7	1467.4	0.465	0.877	1.080	1.1
sarima_s7-noTransform	1	1143.4	1533.9	0.321	0.811	1.201	1.2
THieF_12wk-noTransform	1	1272.4	1793.6	0.402	0.883	1.336	1.4
COVIDhub-4_week_ensemble	2	1156.2	1694.7	0.663	0.920	0.720	0.8
THieF_6wk-4root	2	1413.2	2029.0	0.420	0.847	0.881	0.9
THieF_ensemble-train3	2	1417.1	2046.9	0.393	0.844	0.883	0.9
THieF_ensemble-mean	2	1436.9	2070.8	0.405	0.847	0.895	0.9
THieF_6wk-noTransform	2	1528.5	2177.0	0.469	0.865	0.952	1.0
COVIDhub-baseline	2	1604.9	2113.3	0.742	0.853	1.000	1.0
THieF_12wk-noTransform	2	1858.5	2558.3	0.380	0.853	1.158	1.2
sarima_s7-noTransform	2	1923.1	2536.3	0.408	0.825	1.198	1.2
COVIDhub-4_week_ensemble	3	1670.6	2528.8	0.602	0.900	0.754	0.8
THieF_6wk-4root	3	1867.4	2715.3	0.367	0.840	0.842	0.9
THieF_ensemble-train3	3	1979.6	2845.1	0.351	0.815	0.893	0.9
THieF_6wk-noTransform	3	1989.0	2883.6	0.458	0.853	0.897	0.9
THieF_ensemble-mean	3	1989.0	2860.1	0.389	0.834	0.897	0.9

COVIDhub-baseline	3	2217.0	2993.7	0.712	0.834	1.000	1.0
THieF_12wk-noTransform	3	2369.9	3313.8	0.373	0.865	1.069	1.1
sarima_s7-noTransform	3	2738.4	3581.4	0.458	0.796	1.235	1.1
COVIDhub-4_week_ensemble	4	2173.4	3232.7	0.564	0.891	0.787	0.8
THieF_6wk-4root	4	2330.0	3341.7	0.362	0.795	0.844	0.8
THieF_6wk-noTransform	4	2414.9	3567.4	0.433	0.853	0.875	0.9
THieF_ensemble-train3	4	2501.1	3615.3	0.321	0.795	0.906	0.9
THieF_ensemble-mean	4	2502.2	3613.6	0.375	0.804	0.906	0.9
COVIDhub-baseline	4	2761.4	3782.1	0.670	0.801	1.000	1.0
THieF_12wk-noTransform	4	2852.5	4049.1	0.308	0.856	1.033	1.0
sarima_s7-noTransform	4	3549.4	4630.4	0.452	0.756	1.285	1.2

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rm
COVIDhub-4_week_ensemble	1	16.698	25.106	0.646	0.957	0.698	0.7
THieF_ensemble-train3	1	20.324	30.035	0.547	0.927	0.850	0.9
THieF_ensemble-mean	1	20.763	30.521	0.560	0.927	0.868	0.9
THieF_6wk-4root	1	22.500	32.973	0.491	0.892	0.941	1.0
sarima_s7-noTransform	1	23.110	31.921	0.573	0.888	0.966	1.0
COVIDhub-baseline	1	23.921	31.867	0.797	0.970	1.000	1.0
THieF_6wk-noTransform	1	25.168	36.407	0.525	0.893	1.052	1.1
THieF_12wk-noTransform	1	29.174	42.199	0.492	0.886	1.220	1.3
COVIDhub-4_week_ensemble	2	25.421	37.901	0.658	0.951	0.688	0.7
THieF_6wk-4root	2	30.099	43.938	0.515	0.906	0.815	0.8
THieF_ensemble-train3	2	30.663	44.487	0.565	0.913	0.830	0.8
THieF_ensemble-mean	2	31.037	44.798	0.578	0.913	0.840	0.9
THieF_6wk-noTransform	2	34.995	50.267	0.559	0.885	0.947	1.0
COVIDhub-baseline	2	36.947	49.477	0.804	0.960	1.000	1.0

```
|sarima_s7-noTransform    |          2| 37.319| 50.281|  0.602|  0.873| 1.010| 1.0
|THieF_12wk-noTransform   |          2| 37.354| 54.313|  0.529|  0.883| 1.011| 1.0
|COVIDhub-4_week_ensemble |          3| 35.219| 52.519|  0.645|  0.942| 0.726| 0.7
|THieF_6wk-4root          |          3| 38.976| 56.363|  0.523|  0.899| 0.803| 0.8
|THieF_ensemble-train3    |          3| 41.597| 59.884|  0.565|  0.896| 0.857| 0.9
|THieF_ensemble-mean      |          3| 41.852| 60.071|  0.577|  0.896| 0.862| 0.9
|THieF_6wk-noTransform    |          3| 45.108| 64.825|  0.555|  0.872| 0.929| 0.9
|THieF_12wk-noTransform   |          3| 46.500| 68.160|  0.519|  0.872| 0.958| 1.0
|COVIDhub-baseline        |          3| 48.539| 66.299|  0.787|  0.951| 1.000| 1.0
|sarima_s7-noTransform    |          3| 51.314| 69.074|  0.605|  0.859| 1.057| 1.0
|COVIDhub-4_week_ensemble |          4| 44.394| 65.428|  0.626|  0.930| 0.761| 0.8
|THieF_6wk-4root          |          4| 47.532| 68.482|  0.518|  0.884| 0.815| 0.8
|THieF_ensemble-train3    |          4| 51.037| 73.470|  0.554|  0.886| 0.875| 0.9
|THieF_ensemble-mean      |          4| 51.111| 73.392|  0.565|  0.885| 0.876| 0.9
|THieF_6wk-noTransform    |          4| 53.803| 77.462|  0.546|  0.861| 0.922| 0.9
|THieF_12wk-noTransform   |          4| 54.518| 80.610|  0.501|  0.866| 0.934| 0.9
|COVIDhub-baseline        |          4| 58.355| 81.235|  0.773|  0.947| 1.000| 1.0
|sarima_s7-noTransform    |          4| 63.886| 86.444|  0.605|  0.849| 1.095| 1.0

|model                    | horizon_wk|wave    |    wis|    mae| cov_50| cov_95|
|:------------------------|----------:|:-------|------:|------:|------:|------:|--
|COVIDhub-4_week_ensemble |          1|ba4_ba5 |  277.8|  378.7|  0.890|  1.000| 0
|THieF_6wk-4root          |          1|ba4_ba5 |  350.4|  542.9|  0.605|  0.994| 0
|THieF_ensemble-train3    |          1|ba4_ba5 |  399.1|  685.3|  0.459|  1.000| 0
|THieF_ensemble-mean      |          1|ba4_ba5 |  433.6|  740.3|  0.488|  1.000| 0
|COVIDhub-baseline        |          1|ba4_ba5 |  450.8|  486.9|  0.924|  1.000| 1
|THieF_6wk-noTransform    |          1|ba4_ba5 |  516.5|  822.5|  0.599|  1.000| 1
|sarima_s7-noTransform    |          1|ba4_ba5 |  517.8|  828.6|  0.413|  0.988| 1
```

THieF_12wk-noTransform	1	ba4_ba5	790.7	1319.2	0.349	1.000	1
COVIDhub-4_week_ensemble	2	ba4_ba5	411.4	572.5	0.855	1.000	0
THieF_6wk-4root	2	ba4_ba5	518.8	813.0	0.612	1.000	0
THieF_ensemble-train3	2	ba4_ba5	578.6	966.6	0.491	1.000	0
THieF_ensemble-mean	2	ba4_ba5	621.8	1030.3	0.503	1.000	0
COVIDhub-baseline	2	ba4_ba5	654.2	741.0	0.982	1.000	1
THieF_6wk-noTransform	2	ba4_ba5	727.8	1145.9	0.594	1.000	1
sarima_s7-noTransform	2	ba4_ba5	729.7	1169.6	0.527	1.000	1
THieF_12wk-noTransform	2	ba4_ba5	1012.6	1677.5	0.376	1.000	1
COVIDhub-4_week_ensemble	3	ba4_ba5	563.3	824.7	0.772	1.000	0
THieF_6wk-4root	3	ba4_ba5	683.9	1069.1	0.589	1.000	0
THieF_ensemble-train3	3	ba4_ba5	784.0	1311.5	0.449	1.000	0
COVIDhub-baseline	3	ba4_ba5	825.0	1038.9	1.000	1.000	1
THieF_ensemble-mean	3	ba4_ba5	825.0	1371.3	0.500	1.000	1
THieF_6wk-noTransform	3	ba4_ba5	911.0	1464.4	0.589	1.000	1
sarima_s7-noTransform	3	ba4_ba5	953.5	1514.2	0.570	1.000	1
THieF_12wk-noTransform	3	ba4_ba5	1248.7	2097.1	0.380	1.000	1
COVIDhub-4_week_ensemble	4	ba4_ba5	698.9	1034.3	0.768	1.000	0
THieF_6wk-4root	4	ba4_ba5	801.9	1210.3	0.642	1.000	0
THieF_ensemble-train3	4	ba4_ba5	969.8	1585.1	0.477	1.000	0
COVIDhub-baseline	4	ba4_ba5	972.0	1320.5	1.000	1.000	1
THieF_ensemble-mean	4	ba4_ba5	1010.8	1651.7	0.530	1.000	1
THieF_6wk-noTransform	4	ba4_ba5	1034.0	1631.7	0.609	1.000	1
sarima_s7-noTransform	4	ba4_ba5	1169.6	1821.8	0.603	1.000	1
THieF_12wk-noTransform	4	ba4_ba5	1501.8	2581.8	0.338	1.000	1
THieF_ensemble-mean	1	delta	239.1	305.4	0.857	1.000	0
THieF_ensemble-train3	1	delta	247.9	326.8	0.857	1.000	0

sarima_s7-noTransform	1	delta	270.9	454.3	0.286	1.000	0
THieF_6wk-4root	1	delta	327.6	536.4	0.714	1.000	0
THieF_6wk-noTransform	1	delta	332.5	563.7	0.429	1.000	0
THieF_12wk-noTransform	1	delta	381.8	523.0	1.000	1.000	0
THieF_6wk-noTransform	2	delta	286.5	260.1	1.000	1.000	0
sarima_s7-noTransform	2	delta	290.7	240.0	1.000	1.000	0
THieF_ensemble-mean	2	delta	313.6	215.3	1.000	1.000	0
THieF_ensemble-train3	2	delta	316.2	207.6	1.000	1.000	0
THieF_6wk-4root	2	delta	324.4	280.4	1.000	1.000	0
THieF_12wk-noTransform	2	delta	487.9	404.7	1.000	1.000	0
THieF_6wk-noTransform	3	delta	396.0	511.1	0.857	1.000	0
sarima_s7-noTransform	3	delta	486.3	701.8	1.000	1.000	0
THieF_ensemble-mean	3	delta	509.6	800.2	1.000	1.000	0
THieF_ensemble-train3	3	delta	509.6	797.2	1.000	1.000	0
THieF_6wk-4root	3	delta	547.9	979.9	0.857	1.000	0
THieF_12wk-noTransform	3	delta	585.8	276.3	1.000	1.000	0
THieF_6wk-noTransform	4	delta	530.6	831.2	0.714	1.000	0
THieF_ensemble-mean	4	delta	691.0	1128.0	0.857	1.000	0
THieF_ensemble-train3	4	delta	691.8	1135.1	0.857	1.000	0
THieF_12wk-noTransform	4	delta	710.8	555.6	1.000	1.000	0
sarima_s7-noTransform	4	delta	745.2	1232.6	0.714	1.000	0
THieF_6wk-4root	4	delta	801.6	1496.1	0.429	1.000	0
COVIDhub-4_week_ensemble	1	omicron	1067.9	1595.1	0.460	0.870	0
COVIDhub-baseline	1	omicron	1487.9	2083.4	0.528	0.826	1
THieF_ensemble-train3	1	omicron	1509.9	2136.2	0.240	0.695	3
THieF_ensemble-mean	1	omicron	1517.9	2127.8	0.240	0.695	3
THieF_6wk-noTransform	1	omicron	1632.5	2228.7	0.318	0.734	3

THieF_6wk-4root	1	omicron	1772.1	2437.8	0.260	0.545	3
THieF_12wk-noTransform	1	omicron	1850.9	2381.2	0.435	0.747	4
sarima_s7-noTransform	1	omicron	1881.9	2370.8	0.221	0.604	4
COVIDhub-4_week_ensemble	2	omicron	1919.4	2844.9	0.466	0.839	0
THieF_ensemble-mean	2	omicron	2361.2	3270.0	0.273	0.675	3
THieF_ensemble-train3	2	omicron	2365.4	3287.9	0.260	0.669	3
THieF_6wk-4root	2	omicron	2421.1	3411.4	0.188	0.675	3
THieF_6wk-noTransform	2	omicron	2442.9	3368.8	0.312	0.714	3
COVIDhub-baseline	2	omicron	2579.1	3519.7	0.497	0.702	1
THieF_12wk-noTransform	2	omicron	2827.3	3600.0	0.357	0.688	4
sarima_s7-noTransform	2	omicron	3275.9	4104.9	0.253	0.630	5
COVIDhub-4_week_ensemble	3	omicron	2757.2	4201.2	0.435	0.801	0
THieF_6wk-4root	3	omicron	3141.5	4483.1	0.117	0.669	3
THieF_6wk-noTransform	3	omicron	3167.4	4447.5	0.305	0.695	3
THieF_ensemble-mean	3	omicron	3250.5	4481.1	0.247	0.656	3
THieF_ensemble-train3	3	omicron	3273.1	4511.8	0.221	0.617	3
COVIDhub-baseline	3	omicron	3583.0	4912.1	0.429	0.671	1
THieF_12wk-noTransform	3	omicron	3601.4	4700.2	0.338	0.721	4
sarima_s7-noTransform	3	omicron	4672.1	5833.2	0.318	0.578	5
COVIDhub-4_week_ensemble	4	omicron	3556.3	5294.6	0.373	0.789	0
THieF_6wk-noTransform	4	omicron	3854.4	5589.8	0.247	0.701	3
THieF_6wk-4root	4	omicron	3897.8	5515.5	0.084	0.584	4
THieF_ensemble-mean	4	omicron	4046.9	5650.1	0.201	0.604	4
THieF_ensemble-train3	4	omicron	4084.8	5718.6	0.143	0.584	4
THieF_12wk-noTransform	4	omicron	4274.3	5646.6	0.247	0.708	4
COVIDhub-baseline	4	omicron	4439.6	6090.8	0.360	0.615	1
sarima_s7-noTransform	4	omicron	6010.4	7538.6	0.292	0.506	6

model	horizon_wk	wave	wis	mae	cov_50	cov_95
COVIDhub-4_week_ensemble	1	ba4_ba5	8.911	13.434	0.719	0.991
THieF_6wk-4root	1	ba4_ba5	9.947	14.828	0.618	0.968
THieF_ensemble-train3	1	ba4_ba5	10.477	15.622	0.674	0.988
THieF_ensemble-mean	1	ba4_ba5	11.101	16.471	0.687	0.990
sarima_s7-noTransform	1	ba4_ba5	12.808	19.000	0.676	0.977
THieF_6wk-noTransform	1	ba4_ba5	13.925	20.539	0.696	0.992
COVIDhub-baseline	1	ba4_ba5	14.488	16.595	0.921	1.000
THieF_12wk-noTransform	1	ba4_ba5	18.372	28.320	0.607	0.989
COVIDhub-4_week_ensemble	2	ba4_ba5	11.854	17.637	0.758	0.996
THieF_6wk-4root	2	ba4_ba5	13.279	19.072	0.687	0.988
THieF_ensemble-train3	2	ba4_ba5	14.153	20.344	0.737	0.994
THieF_ensemble-mean	2	ba4_ba5	14.862	21.204	0.752	0.995
sarima_s7-noTransform	2	ba4_ba5	17.464	25.074	0.751	0.988
THieF_6wk-noTransform	2	ba4_ba5	17.922	25.688	0.767	0.996
COVIDhub-baseline	2	ba4_ba5	19.686	21.668	0.968	1.000
THieF_12wk-noTransform	2	ba4_ba5	21.523	32.823	0.681	0.996
COVIDhub-4_week_ensemble	3	ba4_ba5	15.218	22.860	0.763	0.995
THieF_6wk-4root	3	ba4_ba5	16.821	23.305	0.724	0.993
THieF_ensemble-train3	3	ba4_ba5	17.948	25.285	0.764	0.996
THieF_ensemble-mean	3	ba4_ba5	18.676	26.221	0.777	0.997
THieF_6wk-noTransform	3	ba4_ba5	21.594	31.205	0.779	0.998
sarima_s7-noTransform	3	ba4_ba5	22.064	31.343	0.781	0.992
COVIDhub-baseline	3	ba4_ba5	23.932	27.492	0.975	1.000
THieF_12wk-noTransform	3	ba4_ba5	25.377	39.278	0.697	0.997
COVIDhub-4_week_ensemble	4	ba4_ba5	18.259	27.514	0.757	0.995

THieF_6wk-4root	4	ba4_ba5	19.920	26.570	0.751	0.997
THieF_ensemble-train3	4	ba4_ba5	21.500	29.630	0.776	0.998
THieF_ensemble-mean	4	ba4_ba5	22.126	30.413	0.788	0.999
THieF_6wk-noTransform	4	ba4_ba5	24.166	34.327	0.800	0.999
sarima_s7-noTransform	4	ba4_ba5	25.953	36.718	0.803	0.995
COVIDhub-baseline	4	ba4_ba5	27.465	32.803	0.975	1.000
THieF_12wk-noTransform	4	ba4_ba5	29.061	45.413	0.687	0.999
sarima_s7-noTransform	1	delta	10.381	14.692	0.701	0.975
THieF_ensemble-mean	1	delta	11.595	18.759	0.640	0.995
THieF_ensemble-train3	1	delta	11.872	19.295	0.604	0.995
THieF_6wk-4root	1	delta	14.157	22.679	0.511	0.953
THieF_12wk-noTransform	1	delta	14.886	23.515	0.525	0.948
THieF_6wk-noTransform	1	delta	15.571	25.831	0.511	0.970
sarima_s7-noTransform	2	delta	14.637	20.798	0.695	0.981
THieF_ensemble-mean	2	delta	16.327	26.271	0.654	0.986
THieF_ensemble-train3	2	delta	16.517	26.761	0.640	0.984
THieF_6wk-4root	2	delta	17.992	29.882	0.544	0.978
THieF_12wk-noTransform	2	delta	19.349	30.097	0.596	0.951
THieF_6wk-noTransform	2	delta	19.582	32.199	0.621	0.973
sarima_s7-noTransform	3	delta	23.200	35.709	0.596	0.973
THieF_ensemble-mean	3	delta	25.615	41.494	0.580	0.975
THieF_ensemble-train3	3	delta	25.767	41.935	0.569	0.978
THieF_6wk-4root	3	delta	27.031	44.490	0.519	0.986
THieF_12wk-noTransform	3	delta	27.962	44.220	0.508	0.918
THieF_6wk-noTransform	3	delta	28.402	46.470	0.533	0.945
sarima_s7-noTransform	4	delta	30.502	49.254	0.541	0.945
THieF_ensemble-mean	4	delta	31.711	51.117	0.538	0.967

THieF_ensemble-train3	4	delta	31.797	51.456	0.530	0.970
THieF_6wk-4root	4	delta	32.277	52.556	0.519	0.995
THieF_6wk-noTransform	4	delta	34.183	55.441	0.475	0.929
THieF_12wk-noTransform	4	delta	34.383	54.476	0.489	0.904
COVIDhub-4_week_ensemble	1	omicron	25.018	37.574	0.569	0.921
THieF_ensemble-train3	1	omicron	31.706	46.621	0.403	0.856
THieF_ensemble-mean	1	omicron	31.971	46.748	0.414	0.853
COVIDhub-baseline	1	omicron	33.998	48.183	0.664	0.939
sarima_s7-noTransform	1	omicron	35.196	47.136	0.451	0.784
THieF_6wk-4root	1	omicron	36.899	53.706	0.348	0.804
THieF_6wk-noTransform	1	omicron	38.161	54.611	0.335	0.780
THieF_12wk-noTransform	1	omicron	41.888	58.550	0.362	0.768
COVIDhub-4_week_ensemble	2	omicron	39.326	58.668	0.556	0.906
THieF_6wk-4root	2	omicron	48.671	71.220	0.330	0.815
THieF_ensemble-train3	2	omicron	48.996	71.161	0.377	0.823
THieF_ensemble-mean	2	omicron	49.036	70.919	0.389	0.822
THieF_6wk-noTransform	2	omicron	53.988	77.423	0.334	0.762
COVIDhub-baseline	2	omicron	54.637	77.978	0.635	0.919
THieF_12wk-noTransform	2	omicron	55.135	78.439	0.362	0.759
sarima_s7-noTransform	2	omicron	59.624	78.628	0.439	0.745
COVIDhub-4_week_ensemble	3	omicron	54.848	81.624	0.529	0.891
THieF_6wk-4root	3	omicron	62.249	90.820	0.317	0.798
THieF_ensemble-mean	3	omicron	66.369	95.646	0.372	0.788
THieF_ensemble-train3	3	omicron	66.579	96.197	0.360	0.790
THieF_12wk-noTransform	3	omicron	69.014	98.879	0.336	0.741
THieF_6wk-noTransform	3	omicron	69.993	100.152	0.325	0.740
COVIDhub-baseline	3	omicron	72.688	104.382	0.603	0.903

sarima_s7-noTransform	3	omicron	82.602	109.301	0.425	0.718
COVIDhub-4_week_ensemble	4	omicron	68.905	100.987	0.504	0.869
THieF_6wk-4root	4	omicron	75.300	110.301	0.289	0.767
THieF_12wk-noTransform	4	omicron	80.395	116.309	0.319	0.733
THieF_ensemble-mean	4	omicron	80.412	116.545	0.349	0.770
THieF_ensemble-train3	4	omicron	80.873	117.458	0.337	0.771
THieF_6wk-noTransform	4	omicron	83.755	120.758	0.301	0.723
COVIDhub-baseline	4	omicron	87.327	126.658	0.584	0.897
sarima_s7-noTransform	4	omicron	102.597	136.890	0.414	0.702

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